3). Let $V = \mathbb{R}^n$ (or $\mathbb{C}^n$). Let $S = \{\vec{x}_1, \cdots, \vec{x}_m\}$ where $m > n$, then S is linearly dependent.

Proof: Denote $A = (\vec{x}_1\ \vec{x}_2 \cdots \vec{x}_m) \in \mathbb{R}^{n\times m}$ (or $\mathbb{C}^{n\times m}$)

$\{\vec{x}_1, \cdots, \vec{x}_m\}$ linearly dep. $\Leftrightarrow \exists c_1, \cdots, c_m$ not all zero such that $c_1\vec{x}_1 + c_2\vec{x}_2 + \cdots + c_m\vec{x}_m = \vec{0}$

$\Leftrightarrow A\vec{x} = \vec{0}$ has non-trivial solutions $\Leftrightarrow Null(A) \neq \{\vec{0}\} \Leftrightarrow \dim(Null(A)) > 0 \Leftrightarrow \underline{nullity(A) > 0}$.

proved below!

By rank-nullity thm: $rank(A) + nullity(A) = m$. as $rank(A) \le n$, $nullity(A) \ge m - n > 0$.

4). A set containing the zero vector is automatically linearly dependent.

Jan 10.

Definition: A set $S \subseteq V$ is linearly dependent if S contains a linearly dependent finite set.

Definition: A set $S \subseteq V$ is linearly independent if every finite subset of S is linearly independent.

Basis, Dimension

Definition: A subset $S \subseteq V$ is a basis of V if it satisfies the following two conditions:

1) S is linearly independent, and

2) $V = \operatorname{Span} S$, i.e $\forall \vec{v} \in V$: $\exists \vec{x}_1, \cdots, \vec{x}_n \in S$ and scalars $a_1, \cdots, a_n$ such that $\vec{v} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_n\vec{x}_n$.

Corollary. Let $S = \{\vec{x}_1, \vec{x}_2, \cdots, \vec{x}_n\} \subseteq V$ be a basis of V. Then $\forall \vec{v} \in V$, there exists a unique n-tuples $a_1, \cdots, a_n$ such that $\vec{v} = a_1\vec{x}_1 + \cdots + a_n\vec{x}_n$.

Proof. Suppose there exists scalars $a_1, a_2, \cdots, a_n, b_1, b_2, \cdots, b_n$ such that

$$\vec{x} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_n\vec{x}_n = b_1\vec{x}_1 + b_2\vec{x}_2 + \cdots + b_n\vec{x}_n$$

Then $a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_n\vec{x}_n = b_1\vec{x}_1 + b_2\vec{x}_2 + b_n\vec{x}_n$

$$\Leftrightarrow (a_1-b_1)\vec{x}_1 + (a_2-b_2)\vec{x}_2 + \cdots + (a_n-b_n)\vec{x}_n = \vec{0}$$

Since $\{x_1, x_2, \cdots, x_n\}$ is linearly independent

$$a_1-b_1 = a_2-b_2 = \cdots = a_n-b_n = 0$$

$$\Leftrightarrow a_1=b_1,\ a_2=b_2,\ \cdots,\ a_n=b_n \quad \blacksquare$$

Theorem (Replacement Theorem). Let V be a vector space. Let $G \subseteq V$ such that G contains n vectors and $V = \operatorname{Span} G$. If $L \subseteq V$ is a linearly independent set or an empty set, which contains m vectors. Then

1). $m \leq n$ and

2). There exists a subset $H \subseteq G$ such that H contains exactly $n-m$ vectors and $\operatorname{Span} L \cup H = V$.

<u>Proof</u>. Proof by induction on m.

Base case: When $m=0$, then $L=\phi$. Then 1) $m \leq n$ and take $H=G$.
Then H contains $n-0=n$ vectors and $\operatorname{Span}(L\cup H) = \operatorname{Span} G = V$.

Inductive steps. Suppose the statement is true for any linearly independent set in V containing k vectors. Then we need to prove that the statement is also true for any linearly independent set containing $k+1$ vectors.

i.e. let $L = \{\vec{x}_1, \cdots, \vec{x}_k, \vec{x}_{k+1}\} \subseteq V$ be a linearly independent set in V, then we need to prove: 1) $n \geq k+1 \Leftrightarrow n-k > 0$ and

2) there exists $H \subseteq V$ such that $V = \operatorname{Span}(L\cup H)$ and H contains $n-k-1$ vectors.

Consider $L' = \{\vec{x}_1, \cdots, \vec{x}_k\} \subseteq V$. Then L' is also linearly independent in V and it contains k vectors.

By inductive hypothesis, 1) $n \geq k$. and 2) there exists $H' = \{\vec{y}_1, \cdots, \vec{y}_{n-k}\} \subseteq G$

such that $V = \text{span}(L' \cup H') = \text{span}\{\vec{x}_1, \cdots, \vec{x}_k, \vec{y}_1, \cdots, \vec{y}_{n-k}\}$

As $\{\vec{x}_1, \cdots, \vec{x}_k, \vec{y}_1, \cdots, \vec{y}_{n-k}\}$ is a spanning set of V, and $\vec{x}_{k+1} \in V$,

$\exists\, c_1, \cdots, c_k, d_1, \cdots, d_{n-k}$ $\quad \vec{x}_{k+1} = c_1\vec{x}_1 + \cdots + c_k\vec{x}_k + d_1\vec{y}_1 + \cdots + d_{n-k}\vec{y}_{n-k}$ (*).

1) Proof of "$n-k > 0$": proof by contradiction: Suppose $n-k \leq 0$

as $n-k \geq 0$ (by the inductive hypothesis): $n=k$.

Then $H' = \phi \Rightarrow \vec{x}_{k+1} = c_1\vec{x}_1 + \cdots + c_k\vec{x}_k$

$\Leftrightarrow c_1\vec{x}_1 + c_2\vec{x}_2 + \cdots + c_k\vec{x}_k + (-1)\vec{x}_{k+1} = \vec{0}$

$\Rightarrow \{\vec{x}_1, \vec{x}_2, \cdots, \vec{x}_k, \vec{x}_{k+1}\}$ is linearly dependent, a contradiction.

2) Constructing H such that $H \subseteq G$, $\text{Span } L \cup H = V$ and $|H| = n-k-1$.

Note that in (*), at least one of $d_1, d_2, \cdots, d_{n-k}$ is non-zero

(otherwise, if $d_1 = \cdots = d_{n-k} = 0$, then $\vec{x}_{k+1} = c_1\vec{x}_1 + \cdots + c_k\vec{x}_k$, then again

it implies $\{\vec{x}_1, \cdots, \vec{x}_k, \vec{x}_{k+1}\}$ is linearly dependent, a contradiction)

Without loss of generality, we may assume $d_{n-k} \neq 0$.

Then: $\vec{x}_{k+1} = c_1\vec{x}_1 + \cdots + c_k\vec{x}_k + d_1\vec{y}_1 + \cdots + d_{n-k}\vec{y}_{n-k}$

$$\Leftrightarrow \vec{y}_{n-k} = \frac{1}{d_{n-k}}\left(\vec{x}_{k+1} - c_1\vec{x}_1 - \cdots - c_k\vec{x}_k - d_1\vec{y}_1 - \cdots - d_{n-k-1}\vec{y}_{n-k-1}\right) \quad (\Delta)$$

Let $H = \{\vec{y}_1, \cdots, \vec{y}_{n-k-1}\} \subseteq G$ then H contains exactly $n-k-1$ vectors.

$\forall \vec{x} \in V$: As $\text{span } L' \cup H' = V$, $\exists\, a_1, \cdots, a_k, b_1, \cdots, b_{n-k}$ such that

$$\vec{x} = a_1\vec{x}_1 + a_2\vec{x}_2 + \cdots + a_k\vec{x}_k + b_1\vec{y}_1 + b_2\vec{y}_2 + \cdots + b_{n-k-1}\vec{y}_{n-k-1} + b_{n-k}\vec{y}_{n-k}$$

$\downarrow$ plug in $\triangle$

$$= a_1\vec{x}_1 + \cdots + a_k\vec{x}_k + b_1\vec{y}_1 + \cdots + b_{n-k-1}\vec{y}_{n-k}$$

$$+ b_{n-k}\left[\frac{1}{d_{n-k}}\left(\vec{x}_{k+1} - c_1\vec{x}_1 - \cdots - c_k\vec{x}_k - d_1\vec{y}_1 - \cdots - d_{n-k-1}\vec{y}_{n-k-1}\right)\right]$$

$$= \left(a_1 - \frac{b_{n-k}c_1}{d_{n-k}}\right)\vec{x}_1 + \cdots + \left(a_k - \frac{b_{n-k}c_k}{d_{n-k}}\right)\vec{x}_k + \frac{b_{n-k}}{d_{n-k}}\vec{x}_{k+1} + \left(b_1 - \frac{b_{n-k}d_1}{d_{n-k}}\right)\vec{y}_1 + \cdots + \left(b_{n-k-1} - \frac{b_{n-k}d_{n-k-1}}{d_{n-k}}\right)\vec{y}_{n-k}$$

$$\in \operatorname{Span}\{\underbrace{\vec{x}_1, \ldots, \vec{x}_{k+1}}_{L}, \underbrace{\vec{y}_1, \ldots, \vec{y}_{n-k-1}}_{H}\} = \operatorname{Span} L \cup H$$

$\Rightarrow \quad V \subseteq \operatorname{Span} L \cup H.$

As $L \subseteq V$ and $H \subseteq G$, $\operatorname{Span} L \cup H \subseteq V$

$\Rightarrow \quad V = \operatorname{Span} L \cup H.$ $\blacksquare$

Corollary. Let V be a vector space having a finite basis. Then every basis of V contains the same number of vectors.

<u>Proof</u>: Let B_1 and B_2 be two bases of V with $|B_1| = n_1$ and $|B_2| = n_2$.

Then B_1 and B_2 are both linearly independent and $\operatorname{Span} B_1 = \operatorname{Span} B_2 = V$.

Then applying the replacement theorem by taking $G = B_1$ (as $\operatorname{Span} G = \operatorname{Span} B_1 = V$) and $L = B_2$ (as $L = B_2$ is linearly independent), we can conclude $n_1 \geq n_2$.

applying the replacement theorem again by applying $G = B_2$ (as $\operatorname{Span} G = \operatorname{Span} B_2 = V$) and $L = B_1$ (as $L = B_1$ is linearly independent), we can conclude that $n_2 \geq n_1$

$\Rightarrow \quad n_1 = n_2$ $\blacksquare$.

Definition: Let S be a basis of V.

1). If S is a finite set, then the number of vectors in S is called the dimension of V, and V is called a finite-dimensional vector space.

2). If S contains infinitely many vectors, then V is called an infinite-dimensional vector space.

Corollary. Let V be a finite-dimensional vector space with $\dim V = n$.

(1). Any linearly independent subset of V which contains exactly n vectors is a basis of V.

(2) Any subset of V which spans V is a basis of V.

<u>Proof</u> (1) Let $S = \{\vec{v}_1, \cdots, \vec{v}_n\} \subseteq V$ such that S is linearly independent.

Let B be a basis of V. Then $|B| = n$.

Use the notation in the replacement theorem, take $G = B$, and $L = S$.

Then there exists a set H with $|H| = |G| - |L| = n - n = 0$. such that

$\mathrm{Span}(L \cup H) = V$. Here, as $|H| = 0$, $H = \phi$. and $L = S$

$\Rightarrow \mathrm{Span}(L \cup H) = \mathrm{Span}\, S = V$.

$\Rightarrow$ S is a spanning set of V

$\Rightarrow$ S is a basis of V. // end of Jan 13

(2). Suppose $S = \{\vec{v}_1, \cdots, \vec{v}_n\}$ and $\mathrm{Span}\, S = V$. Then suppose towards a contradictio

that S is linearly dependent. Then there exists $S' \subsetneq S$ such that

$\mathrm{Span}\, S' = V$. Use the notation in the replacement theorem: take $G = S'$

then $|G| < n$, and take L to be any basis of V. then $|L| = n$.